MENTAL HEALTH • ARTIFICIAL INTELLIGENCE • ADOLESCENTS

Faces of the Mind

Can a short face-reading video reveal depression, anxiety, or stress?
A machine-learning study of 11,427 adolescents.

11,427

Adolescents recruited

8,281

Videos kept after quality checks

0.80

Best F1 score achieved

Presented from: Xu, Zhang & Zhang — "Faces of the Mind" (arXiv:2405.20072)

The Idea: Read Mental Health From a Face

Depression and anxiety often show up in tiny facial movements. Instead of a long clinical interview, could a computer just watch a student read a story for 5 minutes and screen for mental health risk?

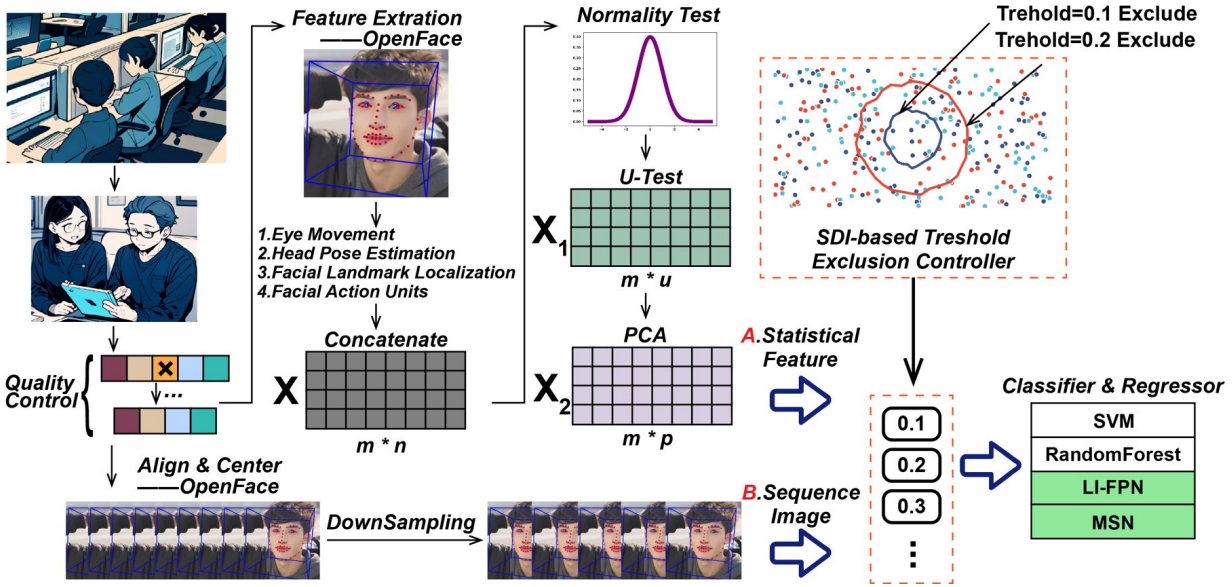
The FACES Study

Who: 11,427 students, ages 10–18, from 11 schools in Jiangsu, China.

What they did: Read a short, neutral story ("The North Wind and the Sun") aloud on a tablet for 5 minutes while the camera recorded their face.

What they filled out: The DASS-21 questionnaire — 21 quick questions rating symptoms of Depression, Anxiety, and Stress.

After quality checks: 8,281 clean videos were kept for analysis (bad lighting, dropped frames, or rushed answers were removed).



How it works (Figure 1): a camera records the face → software (OpenFace) tracks eye movement, head pose, and facial muscle "action units" → those numbers become data → machine-learning models try to predict mental health scores.

The Problem: Same Score, Very Different Symptoms

When the team first trained models on all 8,281 videos, every model scored a weak ~50–60% F1 — barely better than guessing. Why?

Two students, one score

Student A: trouble sleeping + low energy + sad mood
Student B: irritability + poor focus + guilt

Both add up to the exact same total score — so both get labeled “Abnormal.” But their faces likely look nothing alike. Mixing them together confuses the model.

The fix idea — SDI: a new metric that measures how “mixed up” each score group really is.

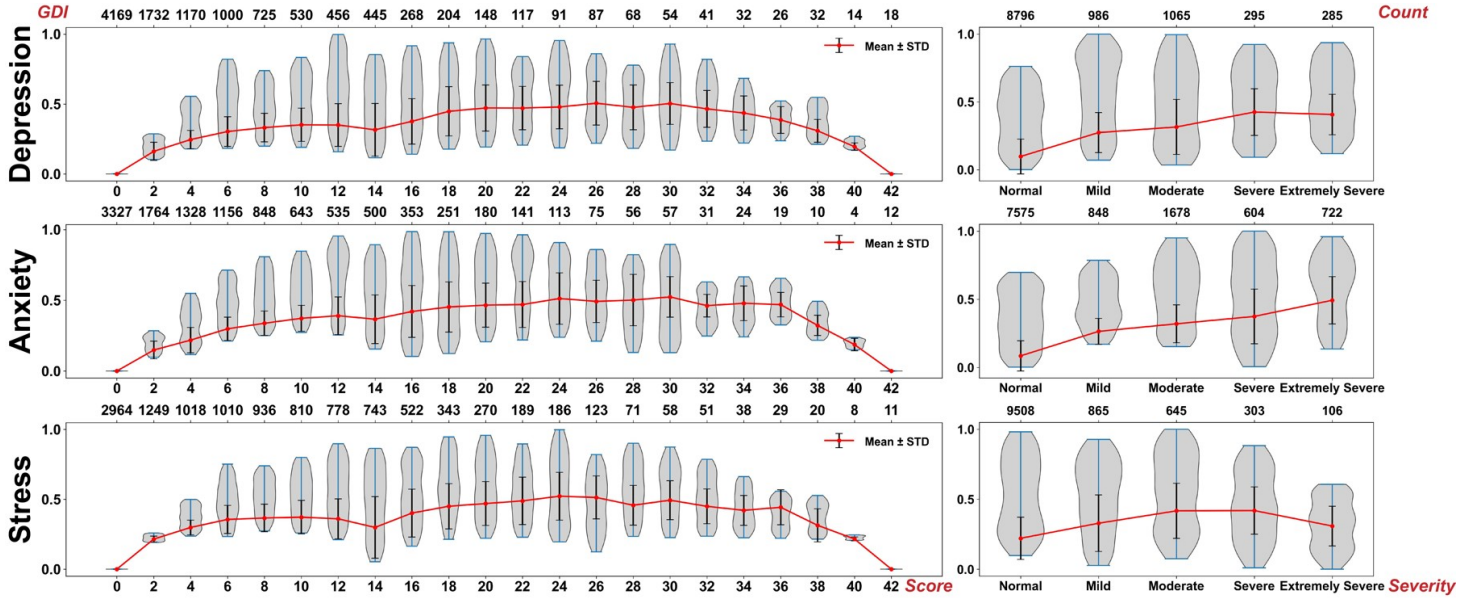


Figure 4: for each possible total score (x-axis), how spread out (“GDI”) are the actual symptom patterns? Spread peaks in the middle scores — exactly the hardest cases to diagnose.

The Fix: Filter Out the Most “Confusing” 10%

Using the Symptom Discrepancy Index (SDI), the researchers found and removed the 10% of students whose symptom pattern least matched their score group — then retrained every model on the cleaner data.

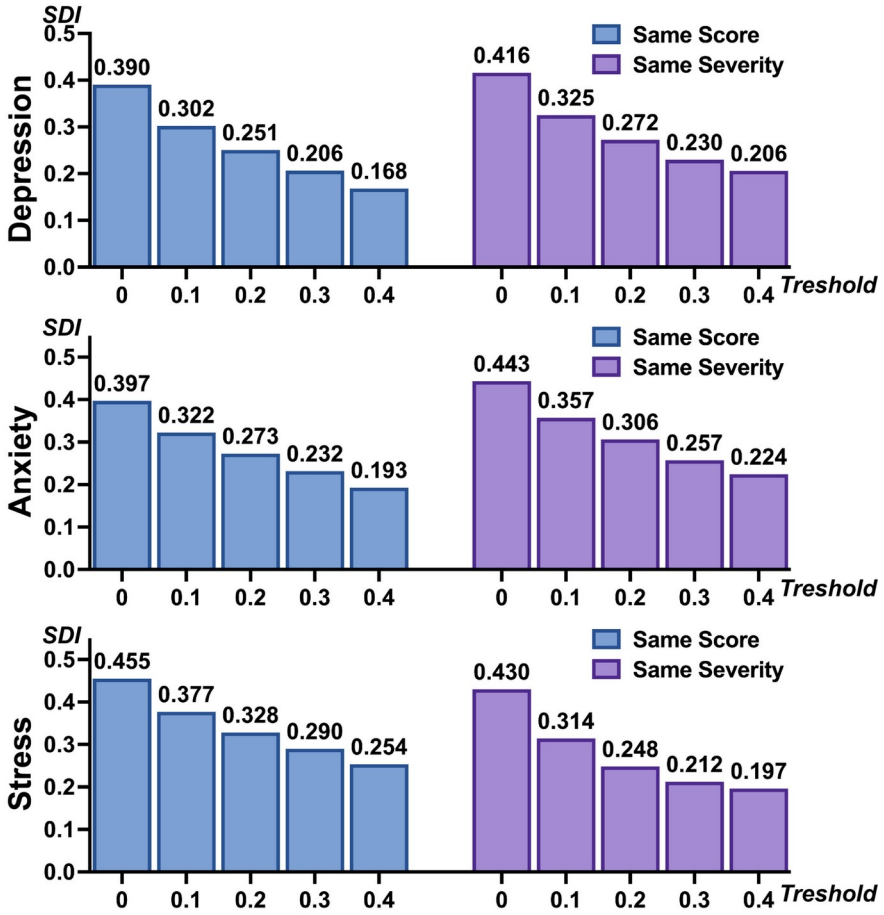
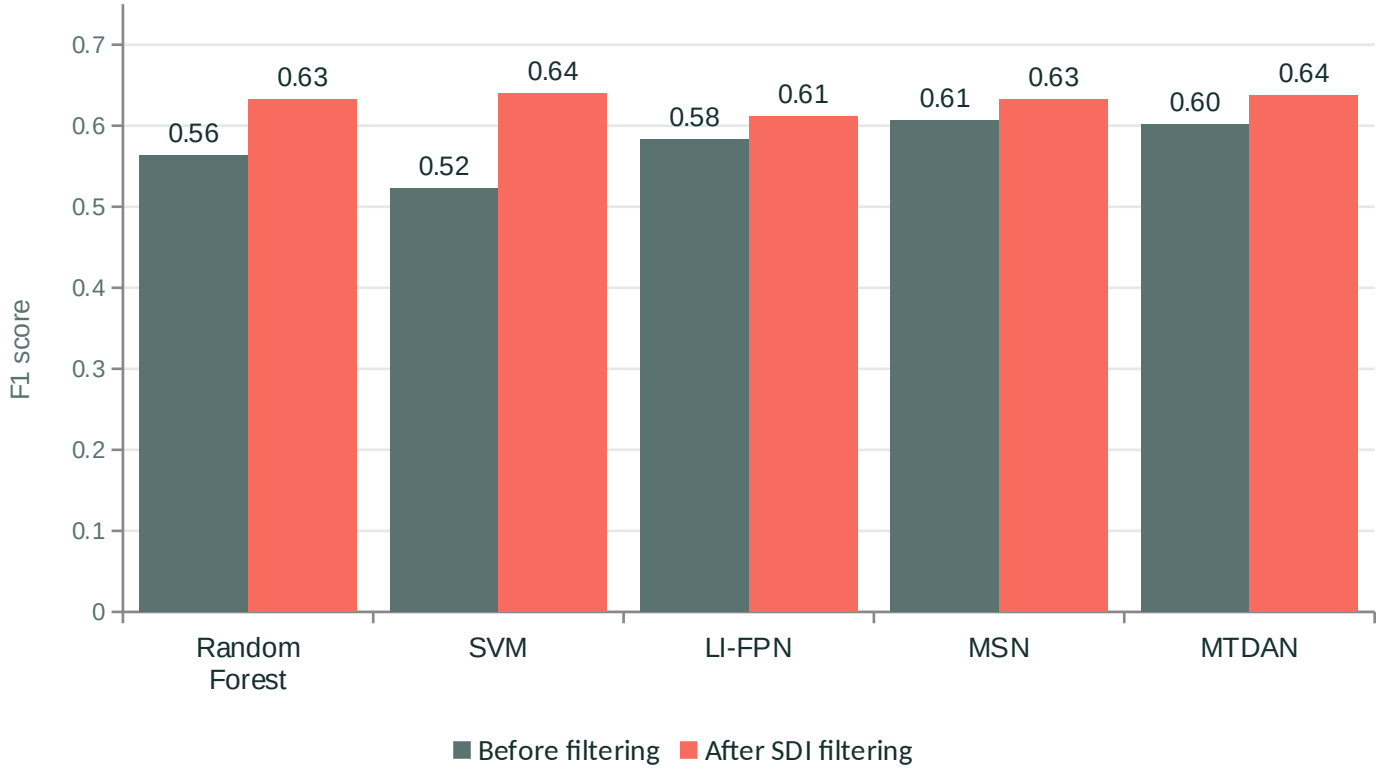


Figure 5: as more “confusing” cases are excluded (x-axis), the leftover heterogeneity (SDI, y-axis) drops — for depression, anxiety, and stress alike.

Every model got better (F1 score, Cluster label, binary task)



Big picture: gains showed up on BOTH the kept data and the removed data — proof this isn’t just deleting “hard” examples, it’s removing genuinely mixed-up labels.

Results & Why It Matters

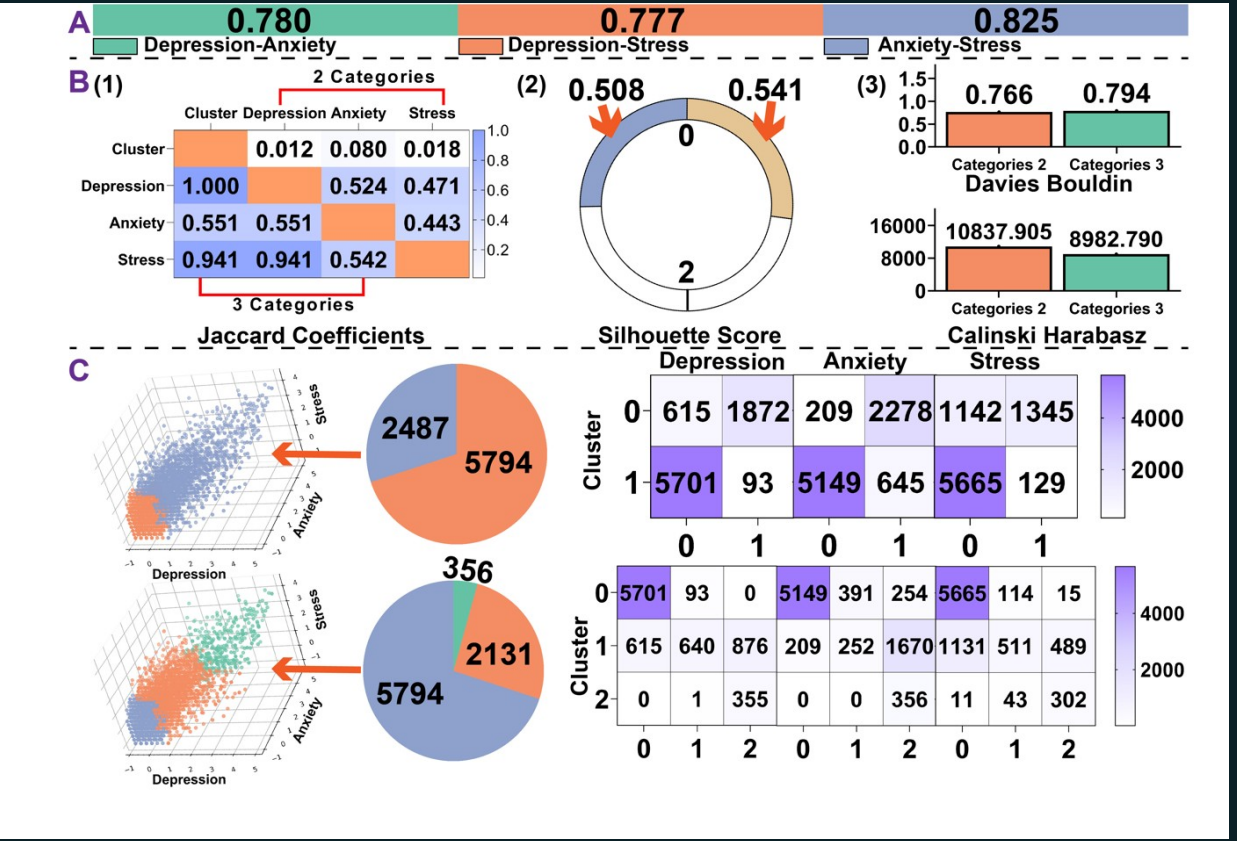


Figure 6: Depression, anxiety, and stress are strongly linked (r = 0.78–0.83). A data-driven clustering step even found a natural 2-group split — roughly "healthy" vs. "at least one elevated symptom."

1 **F1 jumped from ~0.50 to ~0.80**

After SDI-based cleaning, across 5 different ML/deep-learning models.

2 **Regression errors nearly halved**

Predicting exact symptom severity also became far more accurate.

3 **Not a modeling problem — a labeling problem**

The real bottleneck was messy, mixed-up ground-truth labels, not weak algorithms.

4 **Largest dataset of its kind**

11,427 adolescents is ~100x bigger than prior facial mental-health datasets, enabling real-world school screening tools.